

Cobalt: Kagera · Ruvuma · Manyara prospectivity report

Mineral: Cobalt

Region: Kagera

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Executive Summary

Kagera, Ruvuma, and Manyara: Mining Prospectivity Report

Overview

This report explores the use of biogeochemistry as a means for conducting geological prospecting in search of economic grade anomalies of precious metals, base metals, and rare earth elements. The use of randomized soil sampling for geostatistical coefficient calculations and aeromagnetic remote sensing systems in conjunction with biogeochemical prospecting for ferromagnetic elements is examined. The paper concludes that the evidence presented indicates that bioprospecting can be an extremely powerful economic geology tool for conducting rapid and cost effective micro-targeting mineral discovery.

Regional Geological Setting

The report covers Kagera, Ruvuma, and Manyara regions of Tanzania, focusing on the Mesoproterozoic Karagwe-Ankole Belt. Key areas include Kabanga deposit and Ngara district. Detailed geological investigation and drilling are needed to establish resources.

Mineral Potential and Deposit Types

Kabanga Deposit

- **Geological Data:** Provides detailed information about the mineral content, geochemical properties, and forecasting properties of Kabanga deposits.
- **Risk Factors:** High potential for contamination and degradation in the mineral deposit.
- **Recommendations:** Provide specific drilling and geological techniques to improve the accuracy of resource extraction.

Ngara District

- **Geological Data:** Provides detailed information about the mineral content, geochemical properties, and forecasting properties of Ngara district.
- **Risk Factors:** High potential for contamination and degradation in the mineral deposit.
- **Recommendations:** Provide specific drilling and geological techniques to improve the accuracy of resource extraction.

Exploration History and Opportunities



Kabanga Exploration

- **Exploration Camp:** Established in 1993 during the Mururama exploration camp, located approximately 30 km north-west of the current Kabanga camp.
- **Objective:** Conduct exhaustive exploration to find out the potential of its reserves for a subsequent benefit of the mineral.
- **Research Topics:** Includes biogeochemical processing for cobalt recovery, in addition to recovery from secondary sources like batteries and metallurgical waste.

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Infrastructure, Access, and Jurisdiction

Kabanga and Ngara District

- **Geological Inspection:** Conducted by the Mururama exploration camp, located approximately 30 km north-west of the current Kabanga camp.
- **Infrastructure:** Provides detailed geological infrastructure to support mining operations.
- **Access:** Provides access to all necessary resources for mineral extraction.

North Zone Mineral Resources

- **Geological Survey:** Conducted by BHP, at the Project, until the end of 1995.
- **Jurisdiction:** Provides jurisdiction over the North Zone mineral resources.

Risk Factors and Data Limitations

Kabanga Resource Extraction Risk:

- **Risk Level:** High potential for contamination and degradation in the mineral deposit.
- **Impact on Micro-targeting:** Potential benefits for micro-targeting minerals.
- **Recommendations:** Provide specific geological techniques to improve resource extraction.

Ngara Resource Extraction Risk:

- **Risk Level:** High potential for contamination and degradation in the mineral deposit.
- **Impact on Micro-targeting:** Potential benefits for micro-targeting minerals.
- **Recommendations:** Provide specific geological techniques to improve resource extraction.



Recommendations and Next Steps

1. Biogeochemical Processing:

- Conduct extensive biogeochemical processing to refine the mineral content, geochemical properties, and forecasting properties of Kabanga deposits.
- Improve drilling and geological techniques to ensure the accuracy of resource extraction.

2. Resource Extraction and Recovery:

- Conduct exhaustive exploration and drilling to find out the potential of its reserves for a subsequent benefit of the mineral.
- Provide specific drilling and geological techniques to improve the accuracy of resource extraction.

3. Innovative Mining Industry:

- Develop mining industry in Tanzania by reviewing and improving existing practices, such as biogeochemical processing, drilling techniques, and mineral recovery.
- Provide specific mining infrastructure and access to all necessary resources for mineral extraction.

This report covers the entire process of mining prospectivity in Kagera, Ruvuma, and Manyara regions of Tanzania. The report includes detailed geological data, resource extraction and recovery processes, infrastructure, and jurisdictional considerations.

Key findings

- Geological Data: Provides detailed information about the mineral content, geochemical properties, and forecasting properties of Kabanga deposits.
- Risk Factors: High potential for contamination and degradation in the mineral deposit.
- Recommendations: Provide specific drilling and geological techniques to improve the accuracy of resource extraction.
- Geological Data: Provides detailed information about the mineral content, geochemical properties, and forecasting properties of Ngara district.
- Risk Factors: High potential for contamination and degradation in the mineral deposit.
- Recommendations: Provide specific drilling and geological techniques to improve the accuracy of resource extraction.
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- [PDF] stamico boklet final - TISEZA
- "Base Metals Prospecting in Kagera Region, Tanzania Using ...
- Cobalt Metal: Overview of Deposits, Reserves, Processing, and ...
- [PDF] Kabanga 2024 Mineral Resource Update Technical Report Summary
- Development of mining industry in Tanzania: A review

This report is part of the Terra Meta prospectivity catalog. Data and interpretations support exploration planning and due diligence; verify critical decisions with licensed geological survey and field work.

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